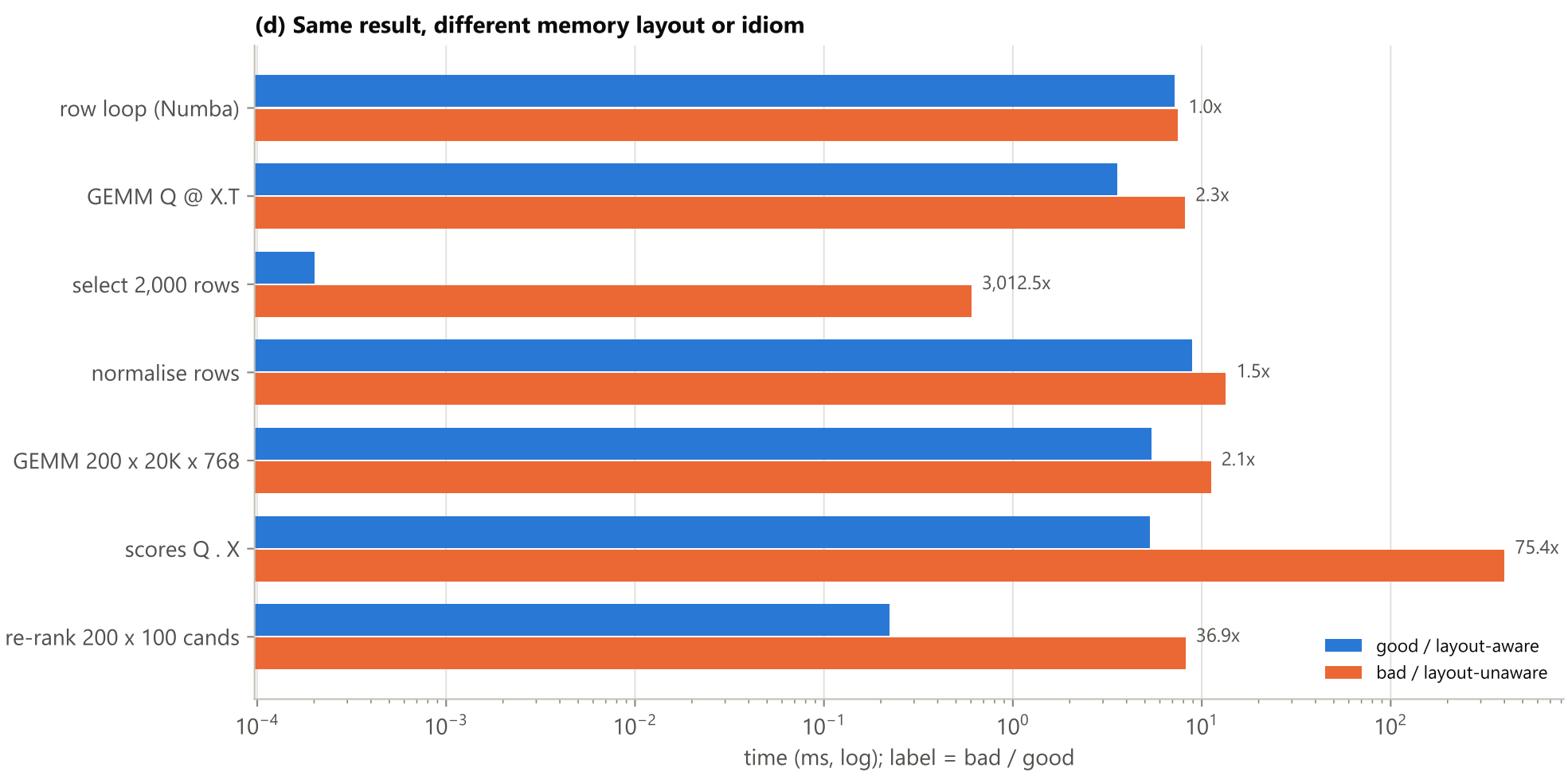
(d) Same result, different memory layout or idiom
row loop (Numba)
GEMM Q @ X.T
select 2,000 rows
normalise rows
GEMM 200 x 20K x 768
scores Q . X
re-rank 200 x 100 cands
1.0x
2.3x
3,012.5x
1.5x
2.1x
75.4x
36.9x
good / layout-aware
bad / layout-unaware
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
time (ms, log); label = bad / good